SENSOR BASLIGI

Motor yatak govdesine vidali – titresim ve yatak sicakligi

Ana karta izole RS–485 ile baglanir (3–10 m ekranli kablo)

01–guc–mcu

Guc ve MCU – 24 V giris, STM32G071

Dosya: 01–guc–mcu.kicad_sch

02–sensorler

Sensorler – IIS3DWB ivmeolcer, 2 x PT100

Dosya: 02–sensorler.kicad_sch

03–haberlesme

Izole RS–485 – ana karta hat

Dosya: 03–haberlesme.kicad_sch

Melike Atılgan

Sheet: /

File: sensor–basligi.kicad_sch

Title: Sensor Basligi

Size: A3	Date: 2026–10–11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 1/4

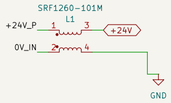
KABLO GIRISI VE KORUMA



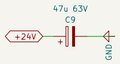


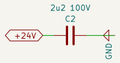


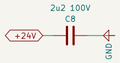










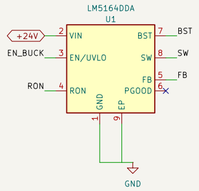








24 V –> 3,3 V SENKRON BUCK



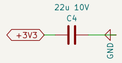


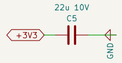


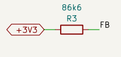






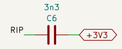






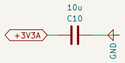






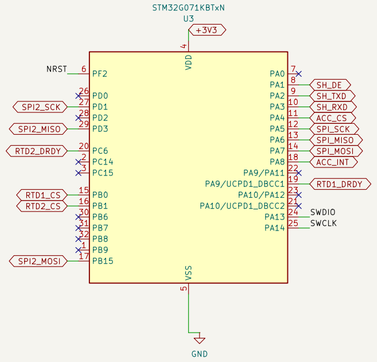


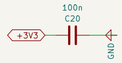


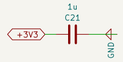


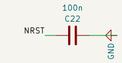


MCU STM32G071KBTxN



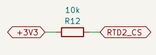


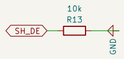


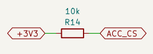


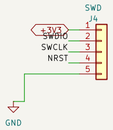












TASARIM NOTLARI

1) KRISTAL YOK. STM32G071 in dahili HSI16 osilatoru +/- %1 kararlilikta; RS-485 115200 baud icin UART toleransi +/- %2, yeterli.

   Titresim ornekleme zamanlamasi ivmeolcerin KENDI saatinden gelir

   (IIS3DWB dahili osilator). MCU saatinden degil. Kristal eklemek basliga iki parca daha eklerdi, olcume katkisi olmazdi.

2) BESLEME 24 V uzerinden kabloyla gelir ve dogrudan 3,3 V a indirilir.

   3,3 V yuku (~0,2 A, cogu izole RS-485) LDO ile verilseydi kapali muhafazada isinma PT100 olcumunu bozardi.

   BUCK (LM5164): Vout = 1.2 V x (1 + 86k6/49k9) = 3,28 V (+0,03 V dalgalanma agi):

   Rron 27k4 –> 301 kHz: 33 uH –> 0,29 A dalgalanma: UVLO 16,9 / 15,7 V.
   Giris: SMBJ33CA en cok 53.3 V ta kirpar, U1 in siniri 100 V. D2 100 V.

   L1 SRF1260-101M: sargilar 1-3 ve 2-4 (uretici foyu).

3) J1-5 BUS_GND: RS-485 in izole referans topragi. Kablo ekrani karta baglanmaz; muhafazanin kablo rakorunda sonlanir.

4) Ortak mod bobini L1 hem besleme hem donus hattinda; motor govdesine monte bir kart, tahrik kaynakli ortak mod gurultusunun icinde durur.

5) BOOT0, G071 de PA14 (SWCLK) uzerindedir; R10 onu asagi ceker.
   R11, R12, R14: resette uc SPI secme hatti yuksek; R13: RS-485 surucusu kapali.

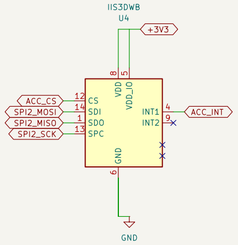
Melike Atilgan

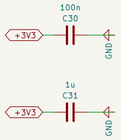
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File: 01-guc-mcu.kicad_sch

Title: Guc ve MCU – 24 V giris, STM32G071KB

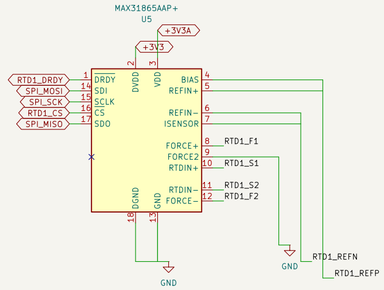
Size: A2	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 2/4

IVMEOLCER IIS3DWB

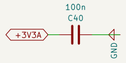


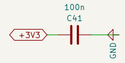


PT100 KANALI 1 (DE yatak)





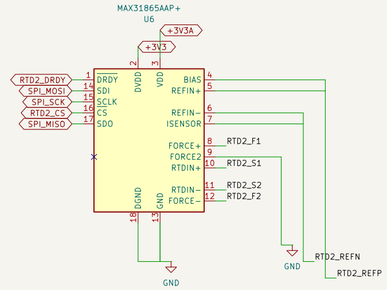




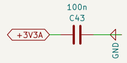


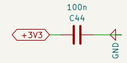


PT100 KANALI 2 (NDE yatak)













TASARIM NOTLARI

1) IIS3DWB durum izleme icin uretilmis bir ivmeolcerdir:
   26,7 kHz orneklеme, 6,3 kHz duz bant, +/− 16 g.
   Genel amacli ivmeolcerlerin bandi (tipik 1−2 kHz) yatak kusur
   frekanslarina ve zarf analizine yetmez.
2) MONTAJ. Baslik yatak govdesine VIDA ile tutturulur. Manyetik taban
   1 kHz ustunde baglanti rezonansi katar ve olcumu bozar; manyetik
   taban yalnizca gezici olcumler icindir.
3) PT100 4 TELLI baglanir. 2 telli baglantida 10 m bakir cift (0,5 mm2)
   yaklasik 0,7 ohm eder; PT100 de bu ~1,8 C hata demektir.
   4 telli baglantida akim FORCE uclarindan, olcum SENSE uclarindan
   yapilir; kablo direnci olcume girmez.
4) Rref = 400 ohm %0,1 10 ppm/C. PT100 icin referans direncin
   kararliligi dogrudan olcum hatasina donusur; sicaklik katsayisi
   direncin TOLERANSINDAN daha onemlidir.
5) BIAS cikisi REFIN+ ya baglidir; Rref, REFIN+ ile REFIN− (ISENSOR)
   arasindadir. Uyarma akimi BIAS −> Rref −> FORCE+ −> RTD −> FORCE−
   yolunu izler. FORCE2 dort telli baglantida topraklanir.
   Siparis kodu MAX31865AAP+ (SSOP−20).
6) RTD girisine 100n: 50 Hz sebeke kaynakli ortak mod gurultusunu
   azaltmak icin; sensor motor govdesinde.

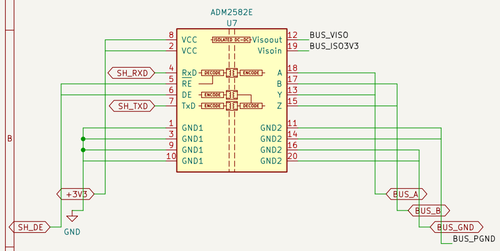
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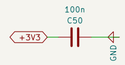
Sheet: /02−sensorler/
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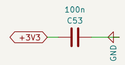
Title: Sensorler − IIS3DWB ve 2 x PT100

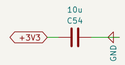
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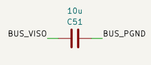
IZOLE RS-485 ADM2582E





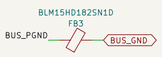


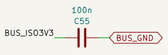
























TASARIM NOTLARI

1) Hat SONUNDAKI cihaz oldugu icin 120 ohm sonlandirma KALICI takilidir.
   Ana kart da hat sonudur; aradaki cihazlarda sonlandirma cikarilir.

2) ADM2582E kendi izole 3,3 V unu uretir. Boylece hat, motor govdesine
   monte basligin kendi toprak referansından da ayrilir.

3) SM712 (SOT-23, uc bacak): iki hatti tek kilifta korur. 1 A darbede
   +19 / −11 V a kirpar; ADM2582E hat siniri −9 / +14 V. Darbe testi yok.

4) -RE ve DE ayni pine bagli: yarim cift yonlu otomatik yon kontrolu.

5) VISOOUT -> VISOIN ve donusturucu topragi (11, 14) -> hat topragi
   (16, 20) ferritle (ADI AN-1349); ferritlerin altinda bakir yok.

Melike Atılgan

Sheet: /03-haberlesme/
File: 03-haberlesme.kicad_sch

Title: Izole RS-485 – ana kart hatti

Size: A2	Date: 2026-10-11	**Rev: C**
KiCad E.D.A. 10.0.6		Id: 4/4